一、Gan Yinbo SCI

1．SCI 论文清单

1 **Nitrogen regulation of root branching**

Walch-Liu, P; Ivanov, II; (...); Forde, BG
17th International Botanical Congress
May 2006 | ANNALS OF BOTANY ▾ 97 (5) , pp.875-881

Background Many plant species can modify their root architecture to enable them to forage for heterogeneously distributed nutrients in the soil. The foraging response normally involves increased proliferation of lateral roots within nutrient-rich soil patches, but much remains to be understood about the signalling mechanisms that enable roots to se ... Show more

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312 Citations
59 References
Related records ⓘ

2 **Integration of cytokinin and gibberellin signalling by *Arabidopsis* transcription factors GIS, ZFP8 and GIS2 in the regulation of epidermal cell fate**

Gan, YB; Liu, C; (...); Broun, P
Jun 1 2007 | DEVELOPMENT ▾ 134 (11) , pp.2073-2081

The effective integration of hormone signals is essential to normal plant growth and development. Gibberellins (GA) and cytokinins act antagonistically in leaf formation and meristem maintenance and GA counteract some of the effects of cytokinins on epidermal differentiation. However, both can stimulate the initiation of defensive epidermal structi ... Show more

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207 Citations
30 References
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3 **Chromium-Induced Reactive Oxygen Species Accumulation by Altering the Enzymatic Antioxidant System and Associated Cytotoxic, Genotoxic, Ultrastructural, and Photosynthetic Changes in Plants**

Wakeel, A; Xu, M and Gan, YB
Feb 2020 | INTERNATIONAL JOURNAL OF MOLECULAR SCIENCES ▾ 21 (3)

Chromium (Cr) is one of the top seven toxic heavy metals, being ranked 21st among the abundantly found metals in the earth's crust. A huge amount of Cr releases from various industries and Cr mines, which is accumulating in the agricultural land, is significantly reducing the crop development, growth, and yield. Chromium mediates phytotoxicity eitl ... Show more

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172 Citations
171 References
Related records

4 **Zinc Finger Protein 6 (ZFP6) regulates trichome initiation by integrating gibberellin and cytokinin signaling in Arabidopsis thaliana**

Zhou, ZJ; Sun, LL; (...); Gan, YB
May 2013 | NEW PHYTOLOGIST ▾ 198 (3) , pp.699-708

The Arabidopsis trichome is a model system for studying cell development, cell differentiation and the cell cycle in plants. Our previous studies have shown that the ZINC FINGER PROTEINS (ZFPS) controls shoot maturation and epidermal cell fate through GA signaling in Arabidopsis. We have identified a novel C2H2 zinc finger protein ZINC FINGER PROTEIN ... Show more

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166 Citations
46 References
Related records

5 **GLABROUS INFLORESCENCE STEMS modulates the regulation by gibberellins of epidermal differentiation and shoot maturation in *Arabidopsis***

Gan, YB; Kumimoto, R; (...); Broun, P
Jun 2006 | PLANT CELL ▾ 18 (6) , pp.1383-1395

As a plant shoot matures, it transitions through a series of growth phases in which successive aerial organs undergo distinct developmental changes. This process of phase change is known to be influenced by gibberellins (GAs). We report the identification of a putative transcription factor, GLABROUS INFLORESCENCE STEMS (GIS), which regul ... Show more

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157 Citations
35 References
Related records

6 **Nutritional regulation of *ANR1* and other root-expressed MADS-box genes in *Arabidopsis thaliana***

Gan, YB; Filleur, S; (...); Forde, BG
Nov 2005 | PLANTA ▾ 222 (4) , pp.730-742

The ANR1 MADS-box gene in Arabidopsis thaliana (L.) Heynh. has previously been identified as a key regulator of lateral root growth in response to signals from external nitrate (NO3-). We have used quantitative real-time PCR to investigate the responsiveness of ANR1 and 11 other root-expressed MADS-box genes to fluctuations in the supply of I ... Show more

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145 Citations
51 References
Related records

7 **Seed priming with zinc oxide nanoparticles downplayed ultrastructural damage and improved photosynthetic apparatus in maize under cobalt stress**

130 Citations
111 References

Salam, A; Khan, AR; (...); Gan, YB

Feb 5 2022 | JOURNAL OF HAZARDOUS MATERIALS ▼ 423

It is widely known that cobalt (Co) stress adversely affects plant growth and biomass accumulation, posing serious threats to crop production and food security. Nanotechnology is an emerging field in crop sciences for its potential in improving crop production and mitigating various stresses. Although there have been several studies reporting the toxic effects of ... Show more

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Related records

8 **Differential control of seed primary dormancy in *Arabidopsis* ecotypes by the transcription factor SPATULA**

126 Citations
52 References

Vaistij, FE; Gan, YB; (...); Graham, IA

Jun 25 2013 | PROCEEDINGS OF THE NATIONAL ACADEMY OF SCIENCES OF THE UNITED STATES OF AMERICA ▼ 110 (26) , pp.10866-10871

Freshly matured seeds exhibit primary dormancy, which prevents germination until environmental conditions are favorable. The establishment of dormancy occurs during seed development and involves both genetic and environmental factors that impact on the ratio of two antagonistic phytohormones: abscisic acid (ABA), which promotes dormancy, and gibb ... Show more

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9 ***Zinc Finger Protein5* Is Required for the Control of Trichome Initiation by Acting Upstream of *Zinc Finger Protein8* in Arabidopsis**

120 Citations
39 References

Zhou, Z.J; An, L.J; (...); Gan, YB

Oct 2011 | PLANT PHYSIOLOGY ▼ 157 (2) , pp.673-682

Arabidopsis (Arabidopsis thaliana) trichome development is a model system for studying cell development, cell differentiation, and the cell cycle. Our previous studies have shown that the GLABROUS INFLORESCENCE STEMS (GIS) family genes, GIS, GIS2, and ZINC FINGER PROTEIN8 (ZFP8), control shoot maturation and epidermal cell fate by integrating gil ... Show more

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10 ***AtEXP2* Is Involved in Seed Germination and Abiotic Stress Response in *Arabidopsis***

115 Citations
71 References

Yan, A; Wu, MJ; (...); Gan, YB

Jan 3 2014 | PLOS ONE ▼ 9 (1)

Expansins are cell wall proteins that promote cell wall loosening by inducing pH-dependent cell wall extension and stress relaxation. Expansins are required in a series of physiological developmental processes in higher plants such as seed germination. Here we identified an Arabidopsis expansin gene AtEXPA2 that is exclusively expressed in ... Show more

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11 **Overexpressing the *ANR1* MADS-Box Gene in Transgenic Plants Provides New Insights into its Role in the Nitrate Regulation of Root Development**

114 Citations
61 References

Gan, YB; Bernreiter, A; (...); Forde, BG

Jun 2012 | PLANT AND CELL PHYSIOLOGY ▼ 53 (6) , pp.1003-1016

The expression of the ANR1 MADS-box gene was manipulated in transgenic plants to investigate its role in the NO3−-dependent regulation of root development in Arabidopsis thaliana. Constitutive overexpression of ANR1 in roots, achieved using GAL4 enhancer trap lines, resulted in more rapid early seedling development, increased lengths and numbe ... Show more

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12 **A DELLA in Disguise: SPATULA Restrains the Growth of the Developing *Arabidopsis* Seedling**

110 Citations
78 References

Josse, EM; Gan, YB; (...); Halliday, KJ

Apr 2011 | PLANT CELL ▼ 23 (4) , pp.1337-1351

The period following seedling emergence is a particularly vulnerable stage in the plant life cycle. In Arabidopsis thaliana, the phytochrome-interacting factor (PIF) subgroup of basic-helix-loop-helix transcription factors has a pivotal role in regulating growth during this early phase, integrating environmental and hormonal signals. We previously shows ... Show more

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13 **The responses of trichome mutants to enhanced ultraviolet-B radiation in *Arabidopsis thaliana***

105 Citations
32 References

Yan, A; Pan, JB; (...); Feng, HY

Aug 1 2012 | JOURNAL OF PHOTOCHEMISTRY AND PHOTOBIOLOGY B-BIOLOGY ▼ 113 , pp.29-35

To gain a better understanding of the protective function of the trichome in Arabidopsis against UV-B radiation, we performed a study using several Arabidopsis trichome mutants (gl1, gis, gis2, zfp8, try82, and gl3), overexpressing trichome positive regulator lines (35S:GIS and 35S:GIS2), and wild-types (WT) under simulated enhanced UV-B radiation condition ... Show more

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14 *GLABROUS INFLORESCENCE STEMS3* (*GIS3*) regulates trichome initiation and development in *Arabidopsis*

Sun, LL; Zhang, AD; [...]; Gan, YB

Apr 2015 | NEW PHYTOLOGIST ▾ 206 (1) , pp.220-230

Arabidopsis trichome formation is an excellent model for studying various aspects of plant cell development and cell differentiation. Our previous works have demonstrated that several C2H2 zinc finger proteins, including GIS, GIS2, ZFP5, ZFP6 and ZFP8, control trichome cell development through GA and cytokinin signalling in Arabidopsis. We ident ... Show more

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102 Citations
41 References
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15 MADS-box Transcription Factor *OsMADS25* Regulates Root Development through Affection of Nitrate Accumulation in Rice

Yu, CY; Liu, YH; [...]; Gan, YB

Aug 10 2015 | PLOS ONE ▾ 10 (8)

MADS-box transcription factors are vital regulators participating in plant growth and development process and the functions of most of them are still unknown. ANR1 was reported to play a key role in controlling lateral root development through nitrate signal in Arabidopsis. OsMADS25 is one of five ANR1-like genes in Oryza Sativa and belongs to the ANR ... Show more

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93 Citations
53 References
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16 A zinc finger protein gene ZFP5 integrates phytohormone signaling to control root hair development in Arabidopsis

An, L.J; Zhou, Z.J; [...]; Gan, YB

Nov 2012 | PLANT JOURNAL ▾ 72 (3) , pp.474-490

Although root hair development in Arabidopsis thaliana has been extensively studied, it remains unknown whether the zinc finger proteins, the largest family of transcription factors in plants, are involved in this process. Here we report that the C2H2 zinc finger protein ZINC FINGER PROTEIN 5 (ZFP5) is a key regulator of root hair initiation and morphogene ... Show more

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87 Citations
59 References
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17 Signaling mechanisms integrating root and shoot responses to changes in the nitrogen supply

Walch-Liu, P; Filleur, S; [...]; Forde, BG

Jan 2005 | PHOTOSYNTHESIS RESEARCH ▾ 83 (2) , pp.239-250

During their life cycle, plants must be able to adapt to wide variations in the supply of soil nitrogen (N). Changes in N availability, and in the relative concentrations of NO3- and NH4+, are known to have profound regulatory effects on the N uptake systems in the root, on C and N metabolism throughout the plant, and on root and shoot morphology. Optimising ... Show more

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87 Citations
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18 Effect of N fertilizer top-dressing at various reproductive stages on growth, N_2 fixation and yield of three soybean (*Glycine max* (L.) Merr.) genotypes

Gan, YB; Stulen, I; [...]; Kuiper, PJC

Jan 20 2003 | FIELD CROPS RESEARCH ▾ 80 (2) , pp.147-155

Soybean (Glycine max (L.) Merr.) is one of the most important food and cash crops in China and a key protein source for the farmers in northern China. Previous experiments in both the field and greenhouse have shown that N-2 fixation alone cannot meet the N requirement for maximizing soybean yield, and that N top-dressing at the flowering stage \ ... Show more

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82 Citations
35 References
Related records

19 Multiple phytohormones promote root hair elongation by regulating a similar set of genes in the root epidermis in Arabidopsis

Zhang, S; Huang, LL; [...]; Gan, YB

Dec 2016 | JOURNAL OF EXPERIMENTAL BOTANY ▾ 67 (22) , pp.6363-6372

Multiple phytohormones, including auxin, ethylene, and cytokinin, play vital roles in regulating cell development in the root epidermis. However, their interactions in specific root hair cell developmental stages are largely unexplored. To bridge this gap, we employed genetic and pharmacological approaches as well as transcriptional analysis in order to d ... Show more

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79 Citations
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20 OsFTIP7 determines auxin-mediated anther dehiscence in rice

Song, S; Chen, Y; [...]; Yu, H

Jul 2018 | NATURE PLANTS ▾ 4 (7) , pp.495-504

Anther dehiscence determines successful sexual reproduction of flowering plants through timely release of pollen grains for pollination and fertilization. Downregulation of auxin levels during pollen mitosis is essential for promoting anther dehiscence along with pollen maturation. How this key transition of auxin levels is regulated in male organs remai ... Show more

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73 Citations
43 References
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21 **C2H2 Zinc Finger Proteins Response to Abiotic Stress in Plants**

Liu, YH; Khan, AR and Gan, YB

Mar 2022 | INTERNATIONAL JOURNAL OF MOLECULAR SCIENCES ▾ 23 (5)

Abiotic stresses have already exhibited the negative effects on crop growth and development, thereby influencing crop quality and yield. Therefore, plants have developed regulatory mechanisms to adopt against such harsh changing environmental conditions. Recent studies have shown that zinc finger protein transcription factors play a crucial role i ... Show more

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97 References
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22 **Low concentrations of nitrate and ammonium stimulate nodulation and N_2 fixation while inhibiting specific nodulation (nodule DW g^{-1} root dry weight) and specific N_2 fixation (N_2 fixed g^{-1} root dry weight) in soybean**

Gan, YB; Stulen, I; (...); Kuiper, PJC

Jan 2004 | PLANT AND SOIL ▾ 258 (1-2) , pp.281-292

Nitrate N is a major inhibitor of the soybean/Bradyrhizobium symbiosis in legumes and although this inhibition has been studied for many years, as yet no consensus has been reached on the specific and quantitative interactions between nitrate and ammonium supply and N-2 fixation. The effect of nitrate and ammonium supply on plant growth, nodi ... Show more

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69 Citations
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23 **OsFTIP1-Mediated Regulation of Florigen Transport in Rice Is Negatively Regulated by the Ubiquitin-Like Domain Kinase OsUbDKγ4**

Song, SY; Chen, Y; (...); Yu, H

Mar 2017 | PLANT CELL ▾ 29 (3) , pp.491-507

Flowering time is a critical agronomic trait that determines successful seed production and adaptation of crop plants. Photoperiodic control of this process in flowering plants is mediated by the long-distance mobile signal called florigen partly encoded by FLOWERING LOCUS T (FT) in Arabidopsis thaliana and its orthologs in other plant species. ... Show more

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24 **Toxicological effects of bisphenol A on growth and antioxidant defense system in *Oryza sativa* as revealed by ultrastructure analysis**

Ali, I; Liu, BH; (...); Gan, YB

Feb 2016 | ECOTOXICOLOGY AND ENVIRONMENTAL SAFETY ▾ 124 , pp.277-284

The present study was conducted to evaluate the effect of bisphenol A (BPA) on rice seedlings grown in a hydroponic system. The obtained results demonstrated that at lower concentration (up to 10 mu M) BPA had some stimulatory effects on the growth of rice seedlings but at higher doses it significantly inhibited seedlings growth. The photosynthetic pigm ... Show more

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65 Citations
58 References
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25 **The effects of *DELLAs* on growth change with developmental stage and brassinosteroid levels**

Lilley, JLS; Gan, YB; (...); Nemhauser, JL

Oct 2013 | PLANT JOURNAL ▾ 76 (1) , pp.165-173

There are two stages in photomorphogenesis. First, seedlings detect light and open their cotyledons. Second, seedlings optimize their light environment by controlled elongation of the seedling stem or hypocotyl. In this study, we used time-lapse imaging to investigate the relationship between the brassinosteroid (BR) and gibberellin (GA) hormones across bo ... Show more

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62 Citations
46 References
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26 **Involvement of ethylene signaling in zinc oxide nanoparticle-mediated biochemical changes in *Arabidopsis thaliana* leaves**

Khan, AR; Wakeel, A; (...); Gan, YB

Jan 1 2019 | ENVIRONMENTAL SCIENCE NANO ▾ 6 (1) , pp.341-355

The growing use of metallic nanoparticles in industry has resulted in their accumulation in agricultural land, which poses a serious threat to the yield and quality of crops worldwide. Various reports showed that plants face environmental stress through hormones; however, the regulation of ethylene under ZnO NP stress is still unknown. In this study, ethr ... Show more

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106 References
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27 **The putative PRC1 RING-finger protein AtRING1A regulates flowering through repressing *MADS AFFECTING FLOWERING* genes in *Arabidopsis***

Shen, LS; Tikong, ZH; (...); Yu, H

Mar 15 2014 | DEVELOPMENT ▾ 141 (6) , pp.1303-1312

Polycomb group proteins play essential roles in the epigenetic control of gene expression in plants and animals. Although some components of Polycomb repressive complex 1 (PRC1)-like complexes have recently been reported in the model plant Arabidopsis, how they contribute to gene repression remains largely unknown. Here we show that a pr ... Show more

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28 **Effects of N management on growth, N_2 fixation and yield of soybean**

Gan, YB; Stulen, I; [...]; Kuiper, P

Feb 2002 | NUTRIENT CYCLING IN AGROECOSYSTEMS ▾ 62 (2) , pp.163-174

Soybean (Glycine max) is one of the most important food and cash crops in China. Although soybean has the capacity to obtain a large proportion of its N from N-2 fixation, it is common farmer's practice to apply an N top dressing to maximize grain yield. A field experiment was conducted to study the effects of N application as urea at various stages during tl ... Show more

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58 Citations
47 References
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29 **Nitrate regulation of lateral root and root hair development in plants**

Liu, BH; Wu, JY; [...]; Gan, YB

4th International Symposium on the Nitrogen Nutrition of Plants

Jul 25 2020 | JOURNAL OF EXPERIMENTAL BOTANY ▾ 71 (15) , pp.4405-4414

Nitrogen (N) is one of the most important macronutrients for plant growth and development. However, the concentration and distribution of N varies in soil due to a variety of environmental factors. In response, higher plants have evolved a developmentally flexible root system to efficiently take up N under N-limited conditions. Over the past ... Show more

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99 References
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30 ***GLABROUS INFLORESCENCE STEMS (GIS)* is Required for Trichome Branching Through Gibberellic Acid Signaling in Arabidopsis**

An, LJ; Zhou, ZJ; [...]; Gan, YB

Feb 2012 | PLANT AND CELL PHYSIOLOGY ▾ 53 (2) , pp.457-468

Cell differentiation generally corresponds to the cell cycle, typically forming a non-dividing cell with a unique differentiated morphology, and Arabidopsis trichome is an excellent model system to study all aspects of cell differentiation. Although gibberellic acid is reported to be involved in trichome branching in Arabidopsis, the mechanism for su ... Show more

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54 Citations
40 References
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31 **The *Arabidopsis* Gene *zinc finger protein 3*(*ZFP3*) Is Involved in Salt Stress and Osmotic Stress Response**

Zhang, AD; Liu, DD; [...]; Gan, YB

Dec 15 2016 | PLOS ONE ▾ 11 (12)

Plants are continuously challenged by various abiotic and biotic stresses. To tide over these adversities, plants evolved intricate regulatory networks to adapt these unfavorable environments. So far, many researchers have clarified the molecular and genetic pathways involved in regulation of stress responses. However, the mechanism through which these re ... Show more

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53 Citations
49 References
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32 **Nano-Priming against Abiotic Stress: A Way Forward towards Sustainable Agriculture**

Salam, A; Afridi, MS; [...]; Gan, YB

Nov 2022 | SUSTAINABILITY ▾ 14 (22)

Agriculture is directly linked to human life, providing food for survival and health. It is threatened by a number of challenges, such as climate change, resource depletion, and abiotic stresses, including heavy metals (HMs), salinity, drought, etc. Various strategies have been employed to palliate the phytotoxic effects of these stressors from the soil-plant : ... Show more

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52 Citations
195 References
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33 **Genetic and molecular regulation by DELLA proteins of trichome development in Arabidopsis**

Gan, Y; Yu, H; [...]; Broun, P

Nov 2007 | PLANT PHYSIOLOGY ▾ 145 (3) , pp.1031-1042

Gibberellins (GA) are known to influence phase change in Arabidopsis (Arabidopsis thaliana) as well as the development of trichomes, which are faithful epidermal markers of shoot maturation. They modulate these developmental programs in part by antagonizing DELLA repressors of growth, GIBBERELLIC ACID INSENSITIVE (GAI) and REPRESSOR OF g1 ... Show more

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34 **Ethylene mediates dichromate-induced oxidative stress and regulation of the enzymatic antioxidant system-related transcriptome in *Arabidopsis thaliana***

Wakeel, A; Ali, I; [...]; Gan, YB

May 2019 | ENVIRONMENTAL AND EXPERIMENTAL BOTANY ▾ 161 , pp.168-179

Dichromate [Cr(VI)] induces phytotoxicity either directly by interfering in a plant's physiological, biochemical, and/or molecular process or indirectly by stimulating reactive oxygen species accumulation (ROS). The underlying mechanism in Cr(VI)-induced ROS mediated phytotoxicity is elusive. In the current study, we investigate the possibility that Cr(VI)-in ... Show more

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49 Citations
69 References
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35 **Ethylene mediates dichromate-induced inhibition of primary root growth by altering *AUX1* expression and auxin accumulation in *Arabidopsis thaliana***

Wakeel, A; Ali, I; [...]; Gan, YB

Jun 2018 | PLANT CELL AND ENVIRONMENT ▾ 41 (6) , pp.1453-1467

The hexavalent form of chromium [Cr(VI)] causes a major reduction in yield and quality of crops worldwide. The root is the first plant organ that interacts with Cr(VI) toxicity, which inhibits primary root elongation, but the underlying mechanisms of this inhibition remain elusive. In this study, we investigate the possibility that Cr(VI) reduces primary root g ... Show more

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36 **AtGIS, a C2H2 zinc-finger transcription factor from Arabidopsis regulates glandular trichome development through GA signaling in tobacco**

Liu, YH; Liu, DQ; [...]; Gan, YB

Jan 29 2017 | BIOCHEMICAL AND BIOPHYSICAL RESEARCH COMMUNICATIONS ▾ 483 (1) , pp.209-215

Glandular trichome is specialized multicellular structures that have capability to synthesize and secrete secondary metabolites and protect plants from biotic and abiotic stresses. Our previous results revealed that the C2H2 zinc-finger transcription factors (GIS) acts upstream of GL3/EGL3-GL1-TTG1 transcriptional activator complex to regulate trichome initi ... Show more

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45 Citations
37 References
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37 **SPATULA Links Daytime Temperature and Plant Growth Rate**

Sidaway-Lee, K; Josse, EM; [...]; Penfield, S

Aug 24 2010 | CURRENT BIOLOGY ▾ 20 (16) , pp.1493-1497

Plants exhibit a wide variety of growth rates that are known to be determined by genetic and environmental factors, and different plants grow optimally at different temperatures [1, 2], indicating that this is a genetically determined character. Moderate decreases in ambient temperature inhibit vegetative growth, but the mechanism is poorly understood ... Show more

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44 Citations
35 References
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38 ***ZFP5* encodes a functionally equivalent *GIS* protein to control trichome initiation**

Zhou, ZJ; An, L.J; [...]; Gan, YB

2012 | PLANT SIGNALING & BEHAVIOR ▾ 7 (1) , pp.28-30

The Arabidopsis thaliana trichome development is a model system for understanding various aspects of plant cell development and differentiation. The C2H2 zinc finger proteins GIS, GIS2, and ZFP8 play important roles in controlling trichome initiation. In our recent study, we reported that a new C2H2 zinc finger protein, ZINC FINGER PROTEIN 5 (ZFP5), con ... Show more

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43 Citations
21 References
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39 **Progress on trichome development regulated by phytohormone signaling**

An, L.J; Zhou, Z.J; [...]; Gan, YB

2011 | PLANT SIGNALING & BEHAVIOR ▾ 6 (12) , pp.1959-1962

Trichomes are specialized structures that develop from epidermal cells in the aerial parts of plants, and are an excellent model system to study all aspects of cell differentiation including cell fate determination, cell cycle regulation, cell polarity and cell expansion. The development of the trichome is a process of integration of both external signals and er ... Show more

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43 Citations
47 References
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40 ***Zinc finger protein 5* (*ZFP5*) associates with ethylene signaling to regulate the phosphate and potassium deficiency-induced root hair development in *Arabidopsis***

Huang, LL; Jiang, QN; [...]; Gan, YB

Jan 2020 | PLANT MOLECULAR BIOLOGY ▾ 102 (1-2) , pp.143-158

Key Message Zinc finger protein transcription factor ZFP5 positively regulates root hair elongation in response to Pi and potassium deficiency by mainly activating the expression of EIN2 in Arabidopsis. Phosphate (Pi) and potassium (K^+) are major plant nutrients required for plant growth and development, and plants respond to low-nutrient condi ... Show more

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42 Citations
43 References
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41 **SHAGGY-like kinase 12 regulates flowering through mediating CONSTANS stability in *Arabidopsis***

Chen, Y; Song, SY; [...]; Shen, LS

Jun 2020 | SCIENCE ADVANCES ▾ 6 (24)

Photoperiod is a major environmental cue that determines the floral transition from vegetative to reproductive development in flowering plants. Arabidopsis thaliana responds to photoperiodic signals mainly through a central regulator CONSTANS (CO). Although it has been suggested that phosphorylation of CO contributes to its role in photoperiodic cor ... Show more

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41 Citations
50 References
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42 **Biochemical responses and ultrastructural changes in ethylene insensitive mutants of *Arabidopsis thaliana* subjected to bisphenol A exposure**

Ali, I; Jan, M; [...]; Gan, YB

Oct 2017 | ECOTOXICOLOGY AND ENVIRONMENTAL SAFETY ▾ 144 , pp.62-71

Bisphenol A (BPA), an important raw material in plastic industry, has become a serious environmental contaminant due to its wide spread use in different products and increasing release into the environment. BPA is known to cause adverse effects in living organisms including plants. Several studies reported that BPA affects growth and development in plan ... Show more

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41 Citations
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43 **The WRKY6 transcription factor affects seed oil accumulation and alters fatty acid compositions in*Arabidopsis thaliana***

Song, G; Li, XP; [...]; Gan, YB

Aug 2020 | PHYSIOLOGIA PLANTARUM ▾ 169 (4) , pp.812-824

In rapeseed, the oil content of the seed not only supplies energy for seed germination and seedling development but also provides essential dietary nutrients for humans and livestock. Recent studies have revealed that many transcription factors (TFs) regulate the accumulation of fatty acids (FAs) during seed development.WRKY6, a WRKY6 family TF, wa ... Show more

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39 Citations
63 References
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44 ***NbGIS* regulates glandular trichome initiation through GA signaling in tobacco**

Liu, YH; Liu, DD; [...]; Gan, YB

Sep 2018 | PLANT MOLECULAR BIOLOGY ▾ 98 (1-2) , pp.153-167

Key messageA novel gene NbGIS positively regulates glandular trichome initiation through GA Signaling in tobacco. NbMYB123-like regulates glandular trichome initiation by acting downstream of NbGIS in tobacco.Abstract Glandular trichome is a specialized multicellular structure which has capability to synthesize and secrete secondary metabolit ... Show more

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39 Citations
75 References
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45 **Involvement of C2H2 zinc finger proteins in the regulation of epidermal cell fate determination in *Arabidopsis***

Yan, A; Wu, M.J; [...]; Gan, YB

Dec 2014 | JOURNAL OF INTEGRATIVE PLANT BIOLOGY ▾ (植物学报 ▾) | 56 (12) , pp.1112-1117

Cell fate determination is a basic developmental process during the growth of multicellular organisms. Trichomes and root hairs of Arabidopsis are both readily accessible structures originating from the epidermal cells of the aerial tissues and roots respectively, and they serve as excellent models for understanding the molecular mechanisms control ... Show more

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38 Citations
62 References
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46 **Expression patterns of nitrate, phosphate, and sulfate transporters in *Arabidopsis* roots exposed to different nutritional regimes**

Baxo, S.J; An, L.J; [...]; Gan, YB

Sep 2011 | BOTANY BOTANIQUE ▾ 89 (9) , pp.647-653

Nitrate transporter AtNRT2.1 is the key component of the inducible high-affinity nitrate transport system in Arabidopsis thaliana. AtNRT2.1 is primarily expressed in roots and known to be mainly involved under fluctuating nitrogen supply conditions. It is still unknown whether AtNRT2.1 is involved in plant responses to other nutrient fluctuations. In this study, ... Show more

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32 Citations
48 References
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47 **Ethylene mediates CuO NP-induced ultrastructural changes and oxidative stress in *Arabidopsis thaliana* leaves**

Azhar, W; Khan, AR; [...]; Gan, YB

Mar 1 2020 | ENVIRONMENTAL SCIENCE NANO ▾ 7 (3) , pp.938-953

Metallic oxide nanoparticles (NPs) have received great importance due to their unique properties and valuable applications in various fields. The release of nanoparticles into the environment at a large scale has become a serious threat to the ecosystem especially plants. Various reports have illustrated the toxicity of CuO NPs to plants, but the hypothesis ... Show more

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48 **The Effects of Fluctuations in the Nutrient Supply on the Expression of Five Members of the *AGL17* Clade of MADS-Box Genes in Rice**

Yu, CY; Su, S; [...]; Gan, YB

Aug 20 2014 | PLOS ONE ▾ 9 (8)

The ANR1 MADS-box gene in Arabidopsis is a key gene involved in regulating lateral root development in response to the external nitrate supply. There are five ANR1-like genes in Oryza sativa, OsMADS23, OsMADS25, OsMADS27, OsMADS57 and OsMADS61, all of which belong to the AGL17 clade. Here we have investigated the responsiveness of these genes to fluct ... Show more

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49 **Ethylene participates in zinc oxide nanoparticles induced biochemical, molecular and ultrastructural changes in rice seedlings**

Khan, AR; Azhar, W; (...); Gan, YB

Dec 15 2021 | ECOTOXICOLOGY AND ENVIRONMENTAL SAFETY ▾ 226

Nowadays, the applications of engineered nanoparticles (ENPs) have been significantly increased, thereby negatively affecting crop production and ultimately contaminating the food chain worldwide. Zinc oxide nanoparticles (ZnO NPs) induced oxidative stress has been clarified in previous studies. But until now, it has not been investigated that how ethyl ... Show more

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50 **Exploring the Adaptive Responses of Plants to Abiotic Stresses Using Transcriptome Data**

Javaid, MH; Khan, AR; (...); Gan, YB

Feb 2022 | AGRICULTURE BASEL ▾ 12 (2)

In recent decades, global climate change and heavy metal stress have severely affected plant growth and biomass, which has led to a serious threat to food safety and human health. Anthropogenic activities, the rapid pace of urbanization, and the use of modern agricultural technologies have further aggravated environmental conditions, resulting in limit ... Show more

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51 **A rice transcription factor, *OsMADS57*, positively regulates high salinity tolerance in transgenic *Arabidopsis thaliana* and *Oryza sativa* plants**

Wu, JY; Yu, CY; (...); Gan, YB

Nov 2021 | PHYSIOLOGIA PLANTARUM ▾ 173 (3) , pp.1120-1135

MADS-box transcription factors (TFs) play indispensable roles in various aspects of plant growth, development as well as in response to environmental stresses. Several MADS-box genes have been reported to be involved in the salt tolerance in different plant species. However, the role of the transcription factor OsMADS57 under salinity stress is still unkn ... Show more

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24 Citations
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52 **The new insight of auxin functions: transition from seed dormancy to germination and floral opening in plants**

Wu, MJ; Wu, JY and Gan, YB

Jun 2020 | PLANT GROWTH REGULATION ▾ 91 (2) , pp.169-174

In angiosperms, plants go through different transitions to complete a life cycle. These different transition stages are crucial for plant development. External and internal stimuli affect these transitions, including environmental condition and phytohormones. Auxin is one of the classical phytohormones and crucial to plant morphogenesis establishment and va ... Show more

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53 **Melatonin-mediated resistance to copper oxide nanoparticles-induced toxicity by regulating the photosynthetic apparatus, cellular damages and antioxidant defense system in maize seedlings**

Khan, AR; Fan, XM; (...); Gan, YB

Jan 1 2023 | ENVIRONMENTAL POLLUTION ▾ 316

The pollution of nanoparticles (NPs) has linked with severe negative effects on crop productivity. Thus, effective strategies are needed to mitigate the phytotoxicity of NPs. The aim of present study was to evaluate the efficacy of exogenously applied melatonin (MT) in mitigating the toxic effects of copper oxide nanoparticles (CuO NPs) from maize seed ... Show more

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93 References
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54 **Physiological changes in soybean (*Glycine max*) Wuyin9 in response to N and P nutrition**

Gan, YB; Stulen, I; (...); Kuiper, PJC

2002 | ANNALS OF APPLIED BIOLOGY ▾ 140 (3) , pp.319-329

Phosphor-us deficiency is a very common problem in the acid soil of central China. Previous research has shown that starter N and N topdressing at the flowering stage (R1) increased soybean (Glycine max) yield and N 2 fixation (Gan et al., 1997, 2000). However, there is little information available concerning soybean response to P-fertiliser in soybean pr ... Show more

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23 Citations
67 References
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55 **Overexpression of MADS-box transcription factor *OsMADS25* enhances salt stress tolerance in Rice and *Arabidopsis***

Wu, JY; Yu, CY; (...); Gan, YB

Jan 2020 | PLANT GROWTH REGULATION ▾ 90 (1) , pp.163-171

Salinity is the major abiotic stress inhibiting plant growth and yield all over the world. MADS-box family genes play vital roles in various aspects of plant growth and development, such as the development of floral organs and the regulation of flowering time. The MADS-box transcription factor OsMADS25, belongs to ANR1 clade, and is significantly induced by h ... Show more

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56 **Ethylene accelerates copper oxide nanoparticle-induced toxicity at physiological, biochemical, and ultrastructural levels in rice seedlings**

Azhar, W; Khan, AR; (...); Gan, YB

Feb 2023 | ENVIRONMENTAL SCIENCE AND POLLUTION RESEARCH ▾ 30 (10) , pp.26137-26149

The enormous use of metal-based nanoparticles (NPs) in different sectors may result in enhanced accumulation in agricultural soil, which could impose negative effects on crop productivity. Hence, strategies are needed to explore the mechanisms of copper oxide nanoparticle (CuO NP)-induced toxicity in crops. The present study aimed to investigate the leve ... Show more

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57 **A Comparison Between Northern Blotting and Quantitative Real-Time PCR as a Means of Detecting the Nutritional Regulation of Genes Expressed in Roots of *Arabidopsis thaliana***

Gan, YB; Zhou, ZJ; (...); Forde, BG

Mar 20 2011 | AGRICULTURAL SCIENCES IN CHINA ▾ 10 (3) , pp.335-342

Quantitative real-time PCR (qRT-PCR) has become a routine and robust technique for measuring the expression of genes of interest, validating microarray experiments and monitoring biomarkers. However, concerns have been raised over the accuracy of qRT-PCR in China as well as in the rest of the world. We have previously used qRT-PCR to study the resp ... Show more

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58 **The effects of fluctuations in the nutrient supply on the expression of *ANR1* and 11 other MADS box genes in shoots and roots of *Arabidopsis thaliana***

Gan, YB; Zhou, ZJ; (...); Goddard, P

Dec 2010 | BOTANY BOTANIQUE ▾ 88 (12) , pp.1023-1031

We previously reported that the response of ANR1 expression in shoots to nitrogen (N) starvation and resupply was different from its response in roots. However, how the other root-expressed MADS box genes respond to different N fluctuations in the shoot, and how these MADS box genes respond to complete nutrient fluctuations in the root, were unknown. ... Show more

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30 References
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59 **DNA Topoisomerase Iα Affects the Floral Transition**

Gong, XM; Shen, LS; (...); Yu, H

Jun 2017 | PLANT PHYSIOLOGY ▾ 173 (1) , pp.642-654

DNA topoisomerases modulate DNA topology to maintain chromosome superstructure and genome integrity, which is indispensable for DNA replication and RNA transcription. Their function in plant development still remains largely unknown. Here, we report a hitherto unidentified role of Topoisomerase I alpha (TOP1 alpha) in controlling flowe ... Show more

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60 **Zinc Finger Protein 1 (ZFP1) Is Involved in Trichome Initiation in *Arabidopsis thaliana***

Zhang, AD; Liu, YH; (...); Gan, YB

Dec 2020 | AGRICULTURE BASEL ▾ 10 (12)

Arabidopsis trichome is specialized structure that develops from epidermal cells, and is an excellent model system for studying various aspects of plant cell development and cell differentiation. Our previous studies have shown that C2H2 zinc finger protein family genes, including GIS, GIS2, GIS3, ZFP5, ZFP6 and ZFP8, play an important role in controlling tric ... Show more

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61 **Involvement of histone acetylation and deacetylation in regulating auxin responses and associated phenotypic changes in plants**

Wakeel, A; Ali, I; (...); Gan, YB

Jun 2018 | PLANT CELL REPORTS ▾ 37 (1) , pp.51-59

The most recent outcomes about the transcription factors and transcription complexes mediated auxin signaling pathway by the histone acetylation and deacetylation.
The phytohormone auxin, is required to regulate its accumulation spatiotemporally and responses to r ... Show more

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62 **Efficacy of zinc-based nanoparticles in alleviating the abiotic stress in plants: current knowledge and future perspectives**

Khan, AR; Azhar, W; (...); Gan, YB

Nov 2023 | ENVIRONMENTAL SCIENCE AND POLLUTION RESEARCH ▾ 30 (51) , pp.110047-110068

Due to sessile, plants are unable to avoid unfavorable environmental conditions which leads to inducing serious negative effects on plant growth, crop yield, and food safety. Instead, various approaches were employed to mitigate the phytotoxicity of these emerging contaminants from the soil-plant system. However, recent studies based on the exogenous ... Show more

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63 **The *SPATULA* transcription factor regulates seed oil content by controlling seed specific genes in *Arabidopsis thaliana***

Liu, BH; Hua, CM; (...); Gan, YB

May 2017 | PLANT GROWTH REGULATION ▾ 82 (1) , pp.111-121

Seed oil content, as a key economical trait, is influenced by multiple factors during seed development. Recent studies have established that control of seed development and seed oil content are interrelated process, controlled by transcription factors. Previously, a bHLH transcription factor SPATULA was reported to have pleiotropic effect on various org ... Show more

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64 **The synergistic potential of biochar and nanoparticles in phytoremediation and enhancing cadmium tolerance in plants.**

Yasin, Muhammad Umair; Haider, Zulqarnain; (...); Gan, Yinbo

2024-apr | Chemosphere ▾ 354 , pp.141672

Cadmium (Cd) is classified as a heavy metal (HM) and is found into the environment through both natural processes and intensified anthropogenic activities such as industrial operations, mining, disposal of metal-laden waste like batteries, as well as sludge disposal, excessive fertilizer application, and Cd-related product usage. This rising Cd disposal ... Show more

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11 Citations
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65 ***GLABROUS INFLORESCENCE STEMS* regulates trichome branching by genetically interacting with *SIM* in *Arabidopsis***

Sun, LL; Zhou, ZJ; (...); Gan, YB

Jul 2013 | JOURNAL OF ZHEJIANG UNIVERSITY SCIENCE B ▾ (浙江大学学报. B辑, 生物医学与生物技术 ▾) 14 (7) , pp.583-589

Arabidopsis trichomes are large branched single cells that protrude from the epidermis. The first morphological indication of trichome development is an increase in nuclear content resulting from an initial cycle of endoreduplication. Our previous study has shown that the C2H2 zinc finger protein GLABROUS INFLORESCENCE STEMS (GIS) is required for tr ... Show more

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66 **Recent Developments in Rice Molecular Breeding for Tolerance to Heavy Metal Toxicity**

Haider, Z; Ahmad, I; (...); Gan, YB

Apr 25 2023 | AGRICULTURE BASEL ▾ 13 (5)

Heavy metal toxicity generally refers to the negative impact on the environment, humans, and other living organisms caused by exposure to heavy metals (HMs). Heavy metal poisoning is the accumulation of HMs in the soft tissues of organisms in a toxic amount. HMs bind to certain cells and prevent organs from functioning. Symptoms of HM poisoning ca ... Show more

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67 **Cys2/His2-Type Zinc Finger Proteins Regulate Plant Growth and Development**

Liu, YH; Khan, AR; (...); Gan, YB

Sep 3 2022 | CRITICAL REVIEWS IN PLANT SCIENCES ▾ 41 (5) , pp.351-383

Cys2/His2-type Zinc Finger Proteins (C2H2-ZFPs) are known to play vital roles in the regulation of growth and development in plants. Nevertheless, the underlying molecular network is yet to be established. In this review paper, we elaborate on the structure, classification and biological functions of C2H2-ZFPs, and focus on the molecular mechanism ... Show more

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68 **The Multifaceted Role of Jasmonic Acid in Plant Stress Mitigation: An Overview**

Rehman, M; Saeed, MS; (...); Gan, YB

Dec 2023 | PLANTS BASEL ▾ 12 (23)

Plants, being sessile, have developed complex signaling and response mechanisms to cope with biotic and abiotic stressors. Recent investigations have revealed the significant contribution of phytohormones in enabling plants to endure unfavorable conditions. Among these phytohormones, jasmonic acid (JA) and its derivatives, collectively referred t ... Show more

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9 Citations
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69 ***BnaA02.NIP6;1a* encodes a boron transporter required for plant development under boron deficiency in *Brassica napus***

Song, G; Li, XP; (...); Gan, YB

Apr 2021 | PLANT PHYSIOLOGY AND BIOCHEMISTRY ▾ 163 , pp.36-45

Boron (B) is an essential micronutrient for the plant normal growth. In Arabidopsis, NIP6;1 is a boric acid channel required for the proper distribution of boric acid, especially in the nodal regions of shoots. BnaA02.NIP6;1a, a homologous gene of AtNIP6;1 in Brassica napus, was reported to play a key role in B transport activity. However, little is known about th ... Show more

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53 References
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70 *PIL5* represses floral transition in *Arabidopsis* under long day conditions

Wu, MJ; Liu, DQ; (...); Gan, YB

May 15 2018 | BIOCHEMICAL AND BIOPHYSICAL RESEARCH COMMUNICATIONS ▾ 499 (3) , pp.513-518

PHYTOCHROME INTERACING FACTOR 3 LIKE 5 (PIL5), also named PHYTOCHROME INTERACTING FACTOR 1 (PIF1) is an important b-HLH transcription factor in Arabidopsis thaliana. Here we show that mutant of pil5-1 displays early flowering phenotype. We demonstrate that the expressions of the major flowering promoter genes [FLOWERING LOCUS T (FT), S ... Show more

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8 Citations
44 References
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71 Linkage Mapping of Stem Saccharification Digestibility in Rice

Liu, BH; Gómez, LD; (...); McQueen-Mason, SJ

Jul 14 2016 | PLOS ONE ▾ 11 (7)

Rice is the staple food of almost half of the world population, and in excess 90% of it is grown and consumed in Asia, but the disposal of rice straw poses a problem for farmers, who often burn it in the fields, causing health and environmental problems. However, with increased focus on the development of sustainable biofuel production, rice straw has b ... Show more

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72 Interactive mode of biochar-based silicon and iron nanoparticles mitigated Cd-toxicity in maize

Yasin, MU; Hamman, F; (...); Gan, YB

Feb 20 2024 | SCIENCE OF THE TOTAL ENVIRONMENT ▾ 912

Cadmium contamination poses severe environmental and health threats, necessitating effective mitigation strategies. Rice husk biochar (BC) and nanoparticle (NP) treatments are emerging strategies with limited research on their synergistic benefits. This study assesses BC, silicon NPs (nSi), and iron NPs (nFe) modifications (B-nSi, BnFe, and B-nSi-nFe) to n ... Show more

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73 Cobalt stress induces photosynthetic and ultrastructural distortion by disrupting cellular redox homeostasis in maize

Salam, A; Rehman, M; (...); Gan, YB

Jan 2024 | ENVIRONMENTAL AND EXPERIMENTAL BOTANY ▾ 217

Heavy metal contamination of the environment is becoming alarmingly serious with the rapid pace of urbani-zation and unrestricted industrialization. Cobalt (Co) is one of the toxic and non-essential metals for plants; however, it is widely recognized as a beneficial component for numerous enzymes and coenzymes within bio-logical systems. Neverth ... Show more

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74 Detrimental effects of Cd and temperature on rice and functions of microbial community in paddy soils

Munir, R; Jan, MH; (...); Gan, YB

May 1 2023 | ENVIRONMENTAL POLLUTION ▾ 324

Heavy metal (HM) contamination and high environmental temperature (HT) are caused by anthropogenic ac-tivities that negatively impact soil microbial communities and agricultural productivity. Although HM con-taminations have deleterious effects on microbes and plants; there are hardly any reports on the combined effects of HM and HT. He ... Show more

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75 *SPATULA* regulates floral transition and photomorphogenesis in a *PHYTOCHROME B*-dependent manner in *Arabidopsis*

Wu, MJ; Upreti, S; (...); Gan, YB

Sep 18 2018 | BIOCHEMICAL AND BIOPHYSICAL RESEARCH COMMUNICATIONS ▾ 503 (4) , pp.2380-2385

Light is the most important exogenous stimulus regulating plant growth and various developmental processes. Phytochromes, especially PHYTOCHROME B (PHYB) mediates the various light-mediated processes in Arabidopsis. SPATULA (SPT) is an important transcription factor, which has been reported previously to participate in temperature-mediated trans ... Show more

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76 Melatonin alleviated cadmium accumulation and toxicity by modulating phytohormonal balance and antioxidant metabolism in rice.

Munir, Raheel; Yasin, Muhammad Umair; (...); Gan, Yinbo

2024 jan | Chemosphere ▾ 346 , pp.140590

Cadmium (Cd) contamination is an eminent dilemma that jeopardizes global food safety and security, especially through its phytotoxicity in rice; one of the most edible crops. Melatonin (MET) has emerged as a protective phytohormone in stress conditions, but the defensive role and underlying mechanisms of MET against Cd toxicity in rice still re ... Show more

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77 **TOE1/TOE2 Interacting with GIS to Control Trichome Development in *Arabidopsis***

Liu, YH; Yang, SQ; (...); Gan, YB

Apr 2023 | INTERNATIONAL JOURNAL OF MOLECULAR SCIENCES ▾ 24 (7)

Trichomes are common appendages originating and projecting from the epidermal cell layer of most terrestrial plants. They act as a first line of defense and protect plants against different types of adverse environmental factors. GL3/EGL3-GL1-TTG1 transcriptional activator complex and GIS family genes regulate trichome initiation through gibberellin ... Show more

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6 Citations 58 References Related records

78 **RNA-Seq-Based Profiling of *pl* Mutant Reveals Transcriptional Regulation of Anthocyanin Biosynthesis in Rice (*Oryza sativa* L.)**

Xu, RN; Pan, RH; (...); Akhter, D

Sep 2021 | INTERNATIONAL JOURNAL OF MOLECULAR SCIENCES ▾ 22 (18)

Purple-colored leaves in plants attain much interest for their important biological functions and could be a potential source of phenotypic marker in selecting individuals in breeding. The transcriptional profiling helps to precisely identify mechanisms of leaf pigmentation in crop plants. In this study, two genetically unlike rice genotypes, the mutant purpl ... Show more

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79 **Dichromate-induced ethylene biosynthesis, perception, and signaling regulate the variance in root growth inhibition among Shaheen basmati and basmati-385 rice varieties**

Waleed, A; Ali, I; (...); Gan, YB

Jul 2021 | ENVIRONMENTAL SCIENCE AND POLLUTION RESEARCH ▾ 28 (28) , pp.38016-38025

Heavy metals, including a hexavalent form of chromium (Cr(VI)) increasing accumulation in agricultural soil, cause a significant reduction in quality, yield, and growth of rice varieties worldwide. Screening for the selection of tolerant varieties is essential for conventional and molecular breeding. Shaheen basmati (SB) and basmati-385 (B-385) rice varieties, a ... Show more

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6 Citations 36 References Related records

81 **Effect of Bisphenol A-induced Oxidative Stress on the Ultra Structure and Antioxidant Defence System of *Arabidopsis thialiana* Leaves**

Ali, I; Waleed, A; (...); Gan, YB

2018 | POLISH JOURNAL OF ENVIRONMENTAL STUDIES ▾ 27 (3) , pp.967-978

Bisphenol A (BPA) is an emerging environmental poll utant with potentially toxic effects on living organisms. The present study was undertaken to analyze the effects of BPA on the leaves of Arabidopsis thialina by determining the levels of photosynthetic pigments, reactive oxygen species (ROS), membrane lipid peroxidation, and ultrastructural malformat ... Show more

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82 **A peroxisomal cinnamate:CoA ligase-dependent phytohormone metabolic cascade in submerged rice germination**

Wang, YK; Jin, GC; (...); Pan, RH

Jun 3 2024 | DEVELOPMENTAL CELL ▾ 59 (11)

The mechanism underlying the ability of rice to germinate underwater is a largely enigmatic but key research question highly relevant to rice cultivation. Moreover, although rice is known to accumulate salicylic acid (SA), SA biosynthesis is poorly defined, and its role in underwater germination is unknown. It is also unclear whether peroxisomes, organelles ... Show more

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83 **GLABROUS INFLORESCENCE STEMS3 binds to and activates *RHD2* and *RHD4* genes to promote root hair elongation in Arabidopsis**

Huang, LL; Xu, N; (...); Gan, YB

Jan 2024 | PLANT JOURNAL ▾ 117 (1) , pp.92-106

Root hairs are crucial in the uptake of essential nutrients and water in plants. This study showed that a zinc finger protein, GIS3 is involved in root hair growth in Arabidopsis. The loss-of-function gis3 and GIS3 RNAi transgenic line exhibited a significant reduction in root hairs compared to the wild type. The application of 1-aminocyclopropane-1-carboxy ... Show more

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84 ***OsUGE1* is directly targeted by OsGRF6 to regulate root hair length in rice**

Yang, SQ; Xu, N; (...); Gan, YB

May 2023 | THEORETICAL AND APPLIED GENETICS ▾ 136 (5)

Key messageRoot hairs are required for water and nutrient acquisition in plants. Here, we report a novel mechanism that OsUGE1 is negatively controlled by OsGRF6 to regulate root hair elongation in rice.Root hairs are tubular outgrowths generated by the root epidermal cells. They effectively enlarge the soil-root contact area and play essential roles for nutrient ... Show more

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85 **Nuclear translocation of OsMADS25 facilitated by OsNAR2.1 in reponse to nitrate signals promotes rice root growth by targeting *OsMADS27* and OsARF7**

Wu, JF; Yang, SQ; (...); Gan, YB

...MMUNICATIONS ▾ 4 (6)

Nitrate is an important nitrogen source and signaling molecule that regulates plant growth and development. Although several components of the nitrate signaling pathway have been identified, the detailed mechanisms are still unclear. Our previous results showed that OsMADS25 can regulate root development in response to nitrate signals, but the mechani ... Show more

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87 **OsFTIP7 determines auxin-mediated anther dehiscence in rice (vol 4, pg 495, 2018)**

Song, SY; Chen, Y; (...); Yu, H

Dec 2018 | NATURE PLANTS ▾ 4 (12) , pp.1124-1124

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2 Citations
1 Reference
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88 **Genome Sequence of *Clostridium tunisiense* TJ, Isolated from Drain Sediment from a Pesticide Factory**

Sun, LL; Wang, Y; (...); Gan, YB

Dec 2012 | JOURNAL OF BACTERIOLOGY ▾ 194 (24) , pp.6950-6951

Clostridium tunisiense is a Gram-positive, obligate anaerobe that was first isolated in an anaerobic evironment under eutrophication. Here we report the first genome sequence of the Clostridium tunisiense TJ isolated from drain sediment of a pesticide factory in Tianjin, China. The genome is of great importance for both basic and application research.

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2 Citations
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89 **Green-synthesized lignin nanoparticles enhance Zea mays resilience to salt stress by improving antioxidant metabolism and mitigating ultrastructural damage.**

Javaid, Muhammad Haseeb; Chen, Nana; (...); Gan, Yinbo

2024-jul | Chemosphere ▾ 359 , pp.142337

Soil salinity poses a substantial threat to agricultural productivity, resulting in far-reaching consequences. Green-synthesized lignin nanoparticles (LNPs) have emerged as significant biopolymers which effectively promote sustainable crop production and enhance abiotic stress tolerance. However, the defensive role and underlying mechanisms of LNPs agi ... Show more

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1 Citation
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90 ***OsUGE2* Regulates Plant Growth through Affecting ROS Homeostasis and Iron Level in Rice**

Yang, SQ; Chen, NA; (...); Gan, YB

Dec 2024 | RICE ▾ 17 (1)

Background The growth and development of rice (Oryza sativa L.) are affected by multiple factors, such as ROS homeostasis and utilization of iron. Here, we demonstrate that OsUGE2, a gene encoding a UDP-glucose 4-epimerase, controls growth and development by regulating reactive oxygen species (ROS) and iron (Fe) level in rice. Knockout of this g ... Show more

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90 References
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91 **Titanium dioxide nanoparticles seed priming as a remedy for nickel-induced stress in maize through antioxidant enhancement and ultrastructural optimization.**

Rehman, Muhammad; Salam, Abdul; (...); Gan, Yinbo

2024-nov-30 | Journal of environmental management ▾ 373 , pp.123487

Heavy metals (HMs) have emerged as a critical global concern, not only limiting crop productivity but also posing risks to public health. Among them, nickel (Ni) is an essential micronutrient for plant growth; however, it becomes toxic at higher concentrations. Nano-enabled approaches, on the other hand, have emerged as promising eco-friendl ... Show more

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92 **Bioderived carbon quantum dots boost maize growth and photosynthesis by augmenting UV spectrum absorption and carbon assimilation regulatory genes**

Haider, Z; Yang, CY; (...); Gan, YB

Dec 2024 | JOURNAL OF ENVIRONMENTAL CHEMICAL ENGINEERING ▾ 12 (6)

The process of photosynthesis accounts for 90 % of the dry matter produced in crops. To improve photosynthesis, it is more sustainable to employ ecologically safe nanomaterials in order to reduce the risks associated with using metallic or inorganic nanoparticles in plants. This work demonstrates a sustainable method of employing green synthesized ... Show more

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123 References
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94 **Comparative Transcriptome Analysis Provides Insights into Fruit Trichome Development in Peach**

Liu, YH; Xu, M; (...); Gan, YB

Mar 2024 | AGRICULTURE-BASEL 14 (3)

Fruit pubescence (trichome) is an important characteristic and is controlled by a single dominant gene (G/g), resulting in peaches and nectarines. The length and/or density of fruit fuzz varies greatly among different peach cultivars. However, little is known about fruit trichome development in peaches. In this study, significant differences in fruit trichome len ... Show more

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37 References

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96 **OsRbohI Regulates Rice Growth and Development via Jasmonic Acid Signalling**

Qi, JX; Yang, SQ; (...); Gan, YB

Jun 15 2023 | PLANT AND CELL PHYSIOLOGY 64 (6) , pp.686-699

Reactive oxygen species (ROS) are highly reactive molecules, generated by nicotinamide adenine dinucleotide phosphate oxidases encoded by respiratory burst oxidase homologs. The functions of the OsRbohs gene family in rice are diverse and poorly understood. OsRbohI was recently identified as a newly evolved gene in the rice OsRbohs gene family. ... Show more

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97 **Nuclear translocation of OsMADS25 facilitated by OsNAR2.1 in reponse to nitrate signals promotes rice root growth by targeting OsMADS27 and OsARF7**

Wu Junyu; Yang Shuaiqi; (...); Gan Yinbo

2023 | Plant Communications (植物通讯) 4 (6)

Nitrate is an important nitrogen source and signaling molecule that regulates plant growth and development. Although several components of the nitrate signaling pathway have been identified, the detailed mechanisms are still unclear. Our previous results showed that OsMADS25 can regulate root development in response to nitrate signals, but the mechani ... Show more

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73 References

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98 **DNA topoisomerase Iα regulates the floral transition in Arabidopsis**

Gong, XM; Shen, LS; (...); Yu, H

18th International Congress of Developmental Biology

Jul 2017 | MECHANISMS OF DEVELOPMENT 145 , pp.S136-S136

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0 References

99 **The bHLH transcription factor SPT and DELLA proteins act together to regulate cell size of *Arabidopsis* cotyledons in a GA-dependant manner**

Josse, EM; Gan, YB; (...); Halliday, KJ

Annual Meeting of the Society-for-Experimental-Biology

Jun 2009 | COMPARATIVE BIOCHEMISTRY AND PHYSIOLOGY A-MOLECULAR & INTEGRATIVE PHYSIOLOGY 153A (2) , pp.S206-S207

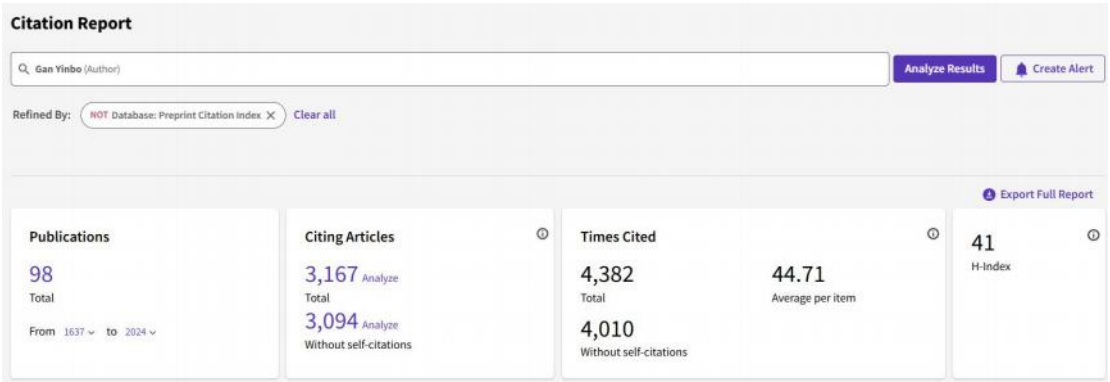
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5 References

Related records

2．Web of Science 上显示的 H 指数为 41

(中文文献和不是甘银波老师的文献并不在前 41 高引用数内）



二、Yinbo Gan SCI

2 **The effect of N fertilizer strategy on N-2 fixation, growth and yield of vegetable soybean**

Yinbo, G; Peoples, MB and Rerkasem, B

Apr 1997 | FIELD CROPS RESEARCH ▾ 51 (3) , pp.221-229

Vegetable soybean (Glycine max) is a lucrative cash crop predominantly grown in Japan, Taiwan, China, Thailand, and Vietnam. Although soybean has the capacity to satisfy a large proportion of its own nitrogen (N) requirements via N-2 fixation, farmers are recommended to apply several side-dressings of fertilizer N during growth in the belief that this will m ... Show more

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51 Citations

27 References

Related records

4 **Interactive mode of biochar-based silicon and iron nanoparticles mitigated Cd-toxicity in maize.**

Muhammad Umair Yasin; Fakhir Hannan; (...); Yinbo Gan

2024 | Science of the Total Environment ▾ 912 , pp.169288

Cadmium contamination poses severe environmental and health threats, necessitating effective mitigation strategies. Rice husk biochar (BC) and nanoparticle (NP) treatments are emerging strategies with limited research on their synergistic benefits. This study assesses BC, silicon NPs (nSi), and iron NPs (nFe) modifications (B-nSi, B-nFe, and B-nSi-nFe) to ı ... Show more

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5 **Melatonin alleviated cadmium accumulation and toxicity by modulating phytohormonal balance and antioxidant metabolism in rice.**

Raheel Munir; Muhammad Umair Yasin; (...); Yinbo Gan

2024 | Chemosphere ▾ 346 , pp.140590

Cadmium (Cd) contamination is an eminent dilemma that jeopardizes global food safety and security, especially through its phytotoxicity in rice; one of the most edible crops. Melatonin (MET) has emerged as a protective phytohormone in stress conditions, but the defensive role and underlying mechanisms of MET against Cd toxicity in rice still re ... Show more

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7 **Detrimental effects of Cd and temperature on rice and functions of microbial community in paddy soils.**

Raheel Munir; Mehmood Jan; (...); Yinbo Gan

2023 | Environmental Pollution ▾ 324 , pp.121371

Heavy metal (HM) contamination and high environmental temperature (HT) are caused by anthropogenic activities that negatively impact soil microbial communities and agricultural productivity. Although HM contaminations have deleterious effects on microbes and plants; there are hardly any reports on the combined effects of HM and HT. Here, we r ... Show more

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0 References

8 **Recent developments in rice molecular breeding for tolerance to heavy metal toxicity.**

Zulqarnain Haider; Irshan Ahmad; (...); Yinbo Gan

2023 | Agriculture ▾ 13 (5) , pp.944

Heavy metal toxicity generally refers to the negative impact on the environment, humans, and other living organisms caused by exposure to heavy metals (HMs). Heavy metal poisoning is the accumulation of HMs in the soft tissues of organisms in a toxic amount. HMs bind to certain cells and prevent organs from functioning. Symptoms of HM poisoning ca ... Show more

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9 **Exploring the adaptive responses of plants to abiotic stresses using transcriptome data.**

Haseeb Javaid, M.; Raza Khan, A.; (...); Yinbo Gan

2022 | Agriculture, 2077-0472 ▾ 12 (2) , pp.211

In recent decades, global climate change and heavy metal stress have severely affected plant growth and biomass, which has led to a serious threat to food safety and human health. Anthropogenic activities, the rapid pace of urbanization, and the use of modern agricultural technologies have further aggravated environmental conditions, resulting in limit ... Show more

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10 **RNA-Seq-based profiling of *pl* mutant reveals transcriptional regulation of anthocyanin biosynthesis in rice (*Oryza sativa* L.).**

Ruonan Xu; Ronghui Pan; (...); Delara Akhter

2021 | International Journal of Molecular Sciences ▾ 22 (18) , pp.9787

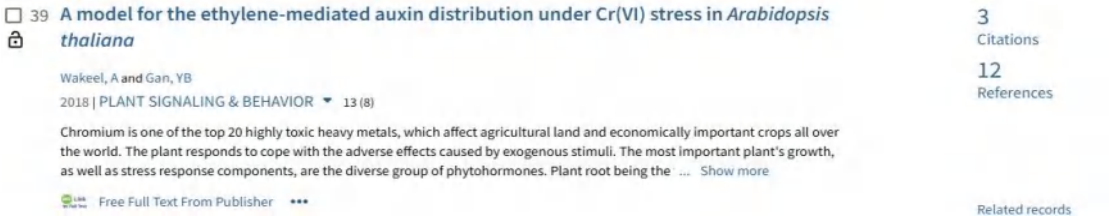
Purple-colored leaves in plants attain much interest for their important biological functions and could be a potential source of phenotypic marker in selecting individuals in breeding. The transcriptional profiling helps to precisely identify mechanisms of leaf pigmentation in crop plants. In this study, two genetically unlike rice genotypes, the mutant purpl ... Show more

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三、Gan Yibo SCI



四、Dr.Ineke Stulen SCI

直接检索 Ineke Stulen 只显示一篇，不过这篇在 Web of Science 上是中文（未包含）



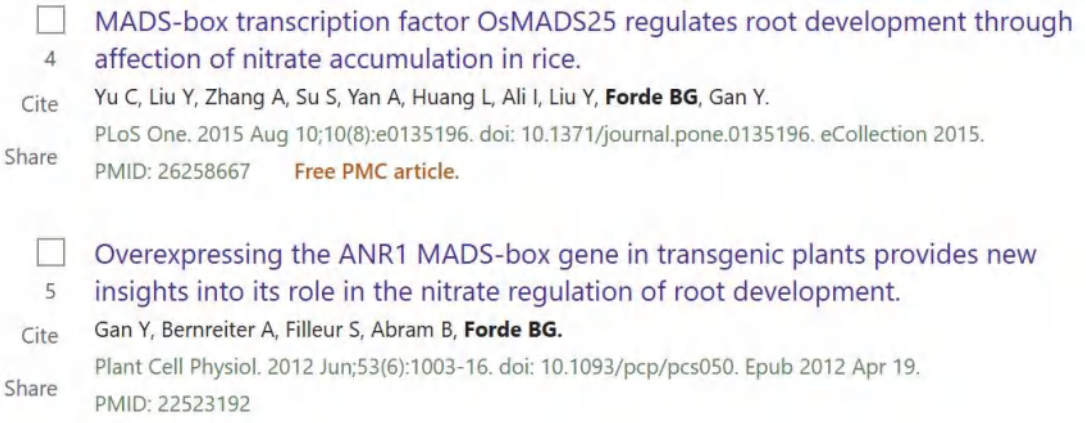
我在寻找时，发现一篇 title 相近的英文论文，但是发表时间不同（已包含）



五、Brian Forde SCI

经过在 PubMed 检索，观察可以看出这几篇现在都已经被 Web of Science 收录（已包含）





六、修正后的 H 指数

Web of Science 上的 H 指数为 41，其余来源中 Yinbo Gan 中有一篇 citations 为 51，而其他论文均不超过 41 或者是已经被 Web of Science 上直接检索 Gan Yinbo report 中包含。

因此修正后的 H 指数应该为 41。

七、问题？

感觉和老师说的有点出入，对于 Yinbo Gan（姓名颠倒）Web of Science 不只有一篇，还有很多新的是 2024 发表的，对于 Ineke Stulen，我在直接检索时发现的是一篇中文文献，但是我在 Gan Yinbo 的检索中发现 Stulen I 应该就是这位博士，对于 Brian Forde 在 Pudmed 的检索中我发现都在 Web of Science 中收录了，而且 Web of Science 中是以 Forde BG 显示的。

所以我觉得应该是 Web of Science 的检索能力和收录能力得到增强，可以更全面更广泛地检索文献，因此和老师记忆中有所偏差可能……